JASWIN CHINTHALA

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PROFESSIONAL PROFILE

First-year Mechatronics Engineering student at the University of Cape Town (UCT) with a focused interest in robotics, autonomous systems, and embedded control. Actively engaged in hardware prototyping at Emi Lab and UCT Racing, complemented by a strong technical background in Python-based automation and computer vision.
Committed to mastering research methodologies to bridge the gap between hardware design and intelligent software

EDUCATION

University of Cape Town 2026–Present

BSc Mechatronics in Electrical Engineering

- Elected EEE1008F Class Representative; facilitating academic communication between students and faculty.
- Focusing on Circuit Analysis, Computer Science, Mathematics, and Physics for Engineering.

Blouberg International School,CapeTown 2021 – 2025

A & AS Levels

- A Levels: Mathematics(A), Physics(B).
- AS Levels: Business Studies(A), Computer Science(B), Chemistry (B), English(B).

TECHNICAL SKILLS

Languages: Python (OpenCV, MediaPipe, Selenium), C Programming.
Hardware: Arduino, ESP32, mmWave Radar Sensors, Circuit Logic, Signal Flow.
Tools: MATLAB, Simulink, Simscape, Microsoft SSO Automation, PyAutoGUI.

ENGINEERING & RESEARCH PROJECTS

E-Scooter Safety System (ESP32/Radar) | Emi Lab, UCT Feb 2026 – Present

- 1 of 8 members developing an ESP32-based safety vest using Bluetooth communication to relay signals to e-scooter controls.
- Integrating mmWave radar sensors for obstacle detection and real-time LED-based direction signaling.

Formula Student Association (FSA) | UCT Racing Feb 2026 – Present

- Collaborating on technical documentation for cooling architecture, specifically X2.4 fan integration.
- Gaining practical exposure to electric drivetrain architecture and battery thermal management.

Independent Research: Hyperspectral Imaging (FreshID) 2026

- Analyzed methodologies for using hyperspectral imaging and AI to detect internal produce degradation.
- Explored SPM-LDA algorithms to correlate visual sub-pixel signatures with chemical senescence indicators.

LEADERSHIP & INVOLVEMENT

Treasurer | Interact Club 2024

- Managed budgets and financial records over R5000+ for community fundraising reaching multiple local organizations.
- 1 of 4 Board Members organizing events, running meetings and participated in various service events

Student Wellness & Leadership | Student Representative Council | Blouberg International School 2024

- Represented student body as 1 of 9 in school governance, contributing to welfare and leadership initiatives.
- Organized mental health and wellness events along with counsellor; mentored cohort of class representatives

Personal Projects

J.A.R.V.I.S: Multi-Modal Automation | Python March 2026

- Architected a multi-threaded system integrating neural wake-word detection and biometric voiceprints.
- Built an FFT-based audio pipeline for clap-pattern recognition and sound-activated triggers.

Automated Portal Entry & Execution March 2026

- Reduced event registration time from 30 seconds to <1 second by high-precision Selenium automation script.
- Overcame cross-origin iframe security barriers by pivoting to screen-coordinate automation via PyAutoGUI.

Skeletal Interface and Gesture Network | Naruto Jutsu March 2026

- Achieved real-time recognition of complex gesture sequences by implementing 21-point landmark tracking and rule-based finger state detection, requiring zero manual ML training.
- Scaling framework to include multi-modal audio cues and an expanded dataset of complex gestures